MILITARY SPECIFICATION

READOUT DRIVER CIRCUIT BOARD: 8623282

1. SCOPE

1.1 This specification covers one type of readout driver printed circuit board, part of the Test Set, Computer Logic Unit: AN/GSM-70.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel: Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-46357	Computer, Gun Direction: M18; Manufacturing Processes and Inspection of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

DRAWINGS

U. S. Army Munitions Command

F8623282	Assembly, Readout Driver
F8623283	Schematic, Electrical

FSC 1220

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PACKAGING DATA SHEET

M14002 Data Sheet, Master Packaging

(Copies of specifications, standards, drawings, and packaging data sheets, required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The readout driver circuit board shall be manufactured in accordance with Drawing F8623282 and drawings pertaining thereto.

3.2 General specifications.- The circuit board shall meet the following requirements of MIL-F-13926, and MIL-C-46357.

3.2.1 MIL-F-13926

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Workmanship

3.2.2 MIL-C-46357

a. Etched circuit assembly
b. Marking
c. Handling

3.3 Environmental service conditions.

3.3.1 Vibration.- The circuit board shall be subjected to a simple harmonic motion having an amplitude of 0.015 inch (0.030 inch total excursion). The frequency shall be varied uniformly from 10 to 55 cycles per second (cps) and return to 10 cycles. This cycle shall be traversed in 1 minute $\pm$ 15 seconds. The motion shall be applied to the circuit board for a period of 10 minutes in each direction of the 3 mutually perpendicular axes (total of 30 minutes), relative to its normal operational position. Subsequent to vibration, the circuit board shall not be damaged or adversely affected and shall meet the requirements of 3.4.

3.3.2 Shock.- The circuit board shall withstand 3 shock impulses in the vertical plane relative to its normal mounting position. The acceleration of the shock impulses shall be 5 gravity units (G's). Pulse duration of each impulse shall be 11 $\pm$ 1 milliseconds. Subsequent to shock the item shall not have been damaged or adversely affected and shall meet the requirements of 3.4.

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3.3.3 Temperatures.

3.3.3.1 Operating.- The circuit board shall meet the requirements of 3.5 while exposed to temperatures of -40°F and +135°F.

3.3.3.2 Storage.- The circuit board shall meet the requirements of 3.4 at a temperature of between 60°F and 90°F after having been exposed to each temperature of -80°F and +160°F.

3.3.4 Humidity.

3.3.4.1 Operating.- The circuit board shall meet the requirements of 3.5 while exposed to an atmosphere with a relative humidity (RH) up to 97% at a temperature of 135°F.

3.3.4.2 Non-operating.- The circuit board shall meet the requirements of 3.4 after exposure to (and when thoroughly dried) an atmosphere with a RH of 100 percent including condensation, for a period of 4 hours.

3.3.5 Fungus resistance.- The circuit board shall show no evidence of damage and shall meet the requirements of 3.4, after being inoculated with the composite spore suspension specified in Specification MIL-F-13926 and incubated at an ambient temperature of between 77°F and 84°F with a RH of 97 ± 2%.

3.3.6 Salt spray.- The circuit board shall show no evidence of physical damage and meet the requirements of 3.4. immediately after exposure to a fine mist of a 5% salt solution concentration. The solution shall have a specific gravity of from 1.0268 to 1.0413 and a pH between 6.5 and 7.2 at a temperature between 93° and 97°F.

3.4 Electrical.- The circuitry shall be in accordance with electrical schematic 8623283.

3.4.1 Readout drivers.

3.4.1.1 True input signal condition.

(a) With the circuits loaded, the base emitter voltage of transistors Q1 through Q21 shall be 0.6 ± 0.3 volts.

(b) With the circuits loaded, the current delivered to each load shall be greater than 0.9 milliamperes.

(c) With no loads connected to the circuits, the output voltage of each circuit shall be less than 10 volts with respect to power ground.

3.4.1.2 False input signal condition.

(a) With the circuits not loaded, the voltage at the emitter of transistors Q1 through Q21 shall be minus 0.2 volts and minus 0.9 volts with respect to power ground.

(b) With the circuits not loaded, the output voltage of each circuit shall be between 85 and 115 volts with respect to power ground.

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3.4.2 AND gate diodes.

3.4.2.1 Any two diodes in a series path to a transistor emitter shall have a combined forward voltage drop of between 0.2 and 0.9 volts.

3.4.2.2 Diodes CR3, CR7, CR11, CR15, CR19, CR23, CR27, CR31, CR35, CR39, CR43, CR47, CR51, CR55, CR6, CR8, CR14, CR16, CR22, CR24, CR30, CR32, CR38, CR40, CR46, CR48, CR54, CR56 shall have a leakage current of less than 6 microamperes when 6 volts back bias is applied.

3.4.2.3 Diodes CR1, CR4, CR9, CR12, CR17, CR20, CR25, CR28, CR33, CR36, CR41, CR44, CR49, CR52, CR57, CR59, CR61, CR63, CR65, CR67, CR69, CR70, CR2, CR5, CR10, CR13, CR18, CR21, CR26, CR29, CR34, CR37, CR42, CR45, CR50, CR53, CR58, CR60, CR62, CR64, CR66, CR68, shall have a leakage current of less than 18 microamperes when 18 volts of back bias is applied.

3.5 Temperature extremes.- The following requirements shall apply at the conditions of 3.3.3.1 and 3.3.4.1.

3.5.1 True state (no load).- With true inputs to each of the logic diodes of an individual driver, the output of the driver shall be less than 50 volts.

3.5.2 True state (load applied).- Under the conditions of 3.5.1, but with a 300k ohm resistor load on the driver, the output shall be less than 80 volts.

3.5.3 False state (no load).- With true inputs to all but one of the logic diodes of an individual driver, the output of the driver shall be greater than 77 volts. This condition shall hold for any combination of one false input and three true inputs to the driver.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Submission of product.- Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

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4.2.2 Examinations and tests.- Except where otherwise specified herein, components and sub-assemblies shall be inspected in accordance with the "Inspection provisions for components and sub-assemblies" and the Sampling Plan Tables contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed in accordance with Tables I, II and III and the test methods and procedures in 4.5. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein.

4.2.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.2.4 Functional tests.- The requirements and tests in Table I shall be inspected on a 100% basis. All examinations and tests shall be conducted at the standard ambient temperature (60° to 90°F).

TABLE I

(100 percent inspection)

	Characteristic	Requirements	Test Procedures
101.	Driver - true input	3.4.1.1	4.5.2.1, 4.5.2.2
102.	Driver - false input	3.4.1.2	4.5.2.3
103.	AND gate diodes - forward	3.4.2.1	4.5.2.4
104.	AND gate diodes - reverse	3.4.2.2, 3.4.2.3	4.5.2.4

4.3 Standard sample.

4.3.1 General.- One circuit board shall be selected at random by a Government representative as a standard sample from each 100 produced. The control sample shall meet the requirements and tests in Table II.

TABLE II

	Characteristic	Requirements	Test Procedures
301.	Fabrication	3.1	Applicable drawings - visual
302.	General specifications	3.2	MIL-F-13926 and MIL-C-46357 - visual

4.3.2 Environmental.- Three units shall be selected at random by a Government representative as a control sample from each 50 produced or from each month's production, whichever occurs first. The control samples shall have met the requirements and tests in Table I and shall then meet the requirements and tests in Table III.

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TABLE III

	Characteristic	Requirements	Test Procedures
303.	Vibration	3.3.1	4.5.1.1
304.	Shock	3.3.2	4.5.1.2
305.	Humidity - operating	3.3.4.1	4.5.1.3.2
306.	Humidity - non-operating	3.3.4.2	4.5.1.3.1
307.	Salt spray	3.3.6	4.5.1.4
308.	Operating temperature	3.3.3.1	4.5.1.5
309.	Storage temperature	3.3.3.2	4.5.1.5

4.3.3 Failure of sample.- Should any one item of a standard sample fail to meet the specified test requirements, acceptance of the represented production lot product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (See 4.2.3).

4.4 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5 and with the "Component Inspection" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.4.1.1 Test equipment.- In conjunction with 4.4.1, the following test equipment shall be utilized in the performance of the applicable tests as specified in 4.5:

NOTE: Items 1 through 5 shall be variable at least through the region 20 percent above and below the nominal value. These items shall have a regulation not greater than plus or minus 2 percent; a peak-to-peak ripple less than 1 percent; and an output impedance less than 2 ohms from dc to 500 KC.

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ITEM NUMBER	DESCRIPTION	
1	Power Supply:	minus 18 volts, nominal current rating greater than or equal to 100 milliamperes.
2	Power Supply:	minus 10 volts, nominal current rating greater than or equal to 100 milliamperes.
3	Power Supply:	minus 3 volts, nominal current rating greater than or equal to 100 milliamperes.
4	Power Supply:	plus 150 volts, nominal current rating greater than or equal to 100 milliamperes.
5	Power Supply:	minus 6 volts, nominal current rating greater than or equal to 100 milliamperes.
6	Voltmeter, D.C. Digital:	A range of at least .001 to 999 volts d.c., or three significant figures. Accuracy of at least ± 0.2% of reading, ± 1 count; input impedance of at least 11 megohms on all ranges.

4.5 Test methods and procedures.

4.5.1 Environmental.

4.5.1.1 Vibration.- The vibration test shall be made on a device with means provided for measuring and controlling the direction of vibration and adjusting the amplitudes and frequencies to keep them within the specified limits of 3.3.1. The measuring instruments of the testing device shall be capable of measuring the vibration amplitude within plus or minus 5 percent, and capable of measuring the frequency within plus or minus 2 percent. The test shall be performed in accordance with the requirements specified in 3.3.1 with the circuit board rigidly mounted in its normal operating position. Subsequent to the vibration test, the unit shall show no damage or defect resulting from the test when examined by visual and tactile means. The unit shall then be inspected as specified in 4.5.1.2 prior to determining compliance with requirements specified under 3.4.

4.5.1.2 Shock.- This test shall be conducted in accordance with Procedure II of the shock test defined in MIL-F-13926 except that the unit shall be shocked in the vertical plane only. The impact (shocks) and "G" value shall be in accordance with 3.3.2. A receiving device, representing the connector receptable that receives the unit in the computer, shall be secured to the horizontal surface of the shock tester. Each end of the receiving device shall be clamped such that the female receiving slot will hold the unit in a vertical plane. The unit will then be inserted into the receiving device and shocked. Subsequent to imposing the number of impacts required, a visual examination shall be made for damages or defects to determine compliance with 3.3.2 and then the unit will be inspected as specified in 4.5.1.3 prior to determining compliance with 3.4 as required by 3.3.2.

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4.5.1.3 Humidity.

4.5.1.3.1 Non-operating.- This test shall be conducted in accordance with the conditions in 3.3.4.2 and the humidity test defined in Procedure I of MIL-F-13926 except that the 4-1/2 hour period shall be reduced to 2 hours, the 6 hour period reduced to 2 hours, and the total test period shall not exceed 5.5 hours per cycle. The unit shall then be thoroughly dried and then subjected to the test in 4.5.1.3.2 prior to meeting the specified requirements of 3.4 as required by 3.3.4.2.

4.5.1.3.2 Operating.- This test shall be conducted in accordance with the conditions in 3.3.4.1 and the humidity test defined in Procedure III of MIL-F-13926 except that the humidity shall be maintained at 97 ± 5 percent for the 24 hour period, and the test shall cease at the end of that period (Method 106 of MIL-STD-202 shall be excluded). Throughout the test the unit shall be operated in accordance with 4.5.3 to determine compliance with 3.3.4.1.

4.5.1.4 Salt spray.- The circuit board shall be subjected to the conditions in 3.3.6 and the salt spray test defined in MIL-STD-202, Condition B. Immediately after exposure, the unit shall be inspected for any physical damage, then tested in accordance with 4.5.2 for compliance with 3.3.6.

4.5.1.5 Temperature (operating and storage).- After being subjected to all environmental tests specified above, the circuit board shall be subjected to Test Procedure I specified in MIL-F-13926, except the temperatures specified in 3.3.3 shall apply. While at the operating temperatures of 3.3.3.1, the unit shall be tested in accordance with 4.5.3 to determine compliance with the requirements of 3.5 as required by 3.3.3.1. The temperature shall then be returned to standard temperature (60° to 90°F) and the unit shall be tested in accordance with 4.5.2 to determine compliance with 3.3.3.2.

4.5.1.6 Fungus.- To determine compliance with 3.3.5, one of the first 5 readout driver circuit boards produced shall be submitted for testing in accordance with the fungus test defined in MIL-F-13926. After approval, subsequent acceptance for fungus requirements may be by certification.

4.5.2 Electrical performance.- Each unit shall be tested in accordance with the test conditions specified herein.

4.5.2.1 True state test (driver loaded).- The following tests shall determine compliance with 3.4.1.1.

(a) Apply a minus 6 volts plus or minus 2 percent to the following pins: 6A, 10A, 20A, 24A, 44A, 51A, 58B, 62A.
(b) Apply the power supply voltages of Table IV.

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TABLE IV

A	B
Power Supply	Pin #
-18V ± 2%	31A
-3V ± 2%	32B
+150V ± 2%	36A
power ground	30A

(c) Connect one end of a 100 kilohm, ± 5%, 1/4 watt resistor to the plus 150 volt power supply. Connect the other end of the resistor to the pins in column A of Table II, one at a time.

(d) Read the voltage at the corresponding pins in column B of Table V with respect to power ground. The voltage shall be between minus 3.3 volts and minus 3.9 volts.

(e) Read the voltage at the pins in column A of Table V with respect to power ground. The voltage shall be less than 60 volts

TABLE V

A	B
1A	5A
2A	3B
3A	9B
7A	6B
11A	15A
14A	9A
18A	14B
21A	20B
25A	22A
28A	27B
29A	29B
38A	35B
42A	36B
45A	43A
49A	45B
52A	50B
56A	54A
59A	57A
63A	61A
64A	65B
65A	62B

4.5.2.2 True state test - (driver not loaded).- This test shall determine compliance with 3.4.1.1

(a) Remove the 100K ohm loads.

(b) Read the voltage at the pins in column A of Table V with respect to power ground. This voltage shall be less than 10 volts.

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4.5.2.3 False state condition.- The following tests shall determine compliance with 3.4.1.2.

(a) Remove all previously made external connections, then ground the following pins: 6A, 10A, 20A, 24A, 44A, 51A, 58B, 62A.

(b) Apply the power supply voltages of Table IV.

(c) Read the voltage at the pins in column B of Table V with respect to power ground. This voltage shall be between minus 0.2 and minus 0.9 volts.

(d) Read the voltage at the pins in column A of Table V with respect to power ground. This voltage shall be between plus 85 and plus 115 volts.

4.5.2.4 Diode tests.- The following tests shall determine compliance with 3.4.2.

(a) Remove all previously made external connections. Then connect minus 18 volts, plus or minus 2 percent to pin 31A.

(b) (1) Apply a minus 6 volts, plus or minus 2 percent through a separate 10 kilohm, plus or minus 1 percent, 1/4 watt resistor to each of the pins in column A of Table VI. At the same time, connect each of the pins in column B to power ground.

(2) Read the voltage at the pins in column A with respect to minus 6 volts. The voltage shall be less than plus 60 millivolts.

(3) Read the voltage at the pins in column B of Table V with respect to power ground. The voltage shall be between minus 0.2 and minus 0.9 volts.

(c) (1) Interchange the connections of 4.5.2.4 (b) (1).

(2) Read the voltage at the pins in column B of Table VI with respect to the minus 6 volts. The voltage shall be less than plus 60 millivolts.

(3) Read the voltage at the pins in column B of Table V with respect to power ground. The voltage shall be between minus 0.2 and minus 0.9 volts.

(4) Remove the connections made in 4.5.2.4 (c) (1).

(d) (1) Apply a minus 6 volts, plus or minus 2 percent through a separate 10 kolohm, plus or minus 1 percent, 1/4 watt resistor to each of the pins in column C of Table VI. At the same time, connect each of the pins in column D of Table VI to power ground.

(2) Read the voltage at the pins in column C of Table VI with respect to the minus 6 volts. The voltage shall be less than plus 60 millivolts.

(3) Read the voltage at the pins in column B of Table V with respect to power ground. The voltage shall be between minus 0.2 and minus 0.9 volts.

(e) (1) Interchange the connections of 4.5.2.4(d) (1).

(2) Read the voltage at the pins in column D of Table VI with respect to the minus 6 volts. The voltage shall be less than plus 60 millivolts.

(3) Read the voltage at the pins in column B of Table V with respect to power ground. The voltage shall be between minus 0.2 volts and minus 0.9 volts.

(4) Remove the connections made in 4.5.2.4 (e) (1).

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(f) Connect the following pins to minus 180 millivolts, plus or minus 5 percent: 30A, 32B, 36A. Connect pin 31A to power ground.

(1) Apply minus 18 volts, plus or minus 2 percent to each of the pins in column A of Table VII. At the same time, apply minus 180 millivolts to the pins in column B of Table VII.

(2) Read the voltage at the pins in column C of Table VII with respect to power ground. The voltage shall be more positive than minus 180 millivolts.

(3) Interchange the connections made in 4.5.2.4 (f) (1).

(4) Read the voltage at the pins in column C of Table VII with respect to power ground. The voltage shall be more positive than minus 180 millivolts.

TABLE VI

A	B	C	D
4B	6A	11B	13A
8B	10A	15B	17A
18B	20A	25B	27A
22B	24A	39B	41A
42B	44A	46B	48A
49B	51A	53B	55A
56B	58B		
60B	62A		

TABLE VII

A	B	C
37B	43B	35B
51B	38B	36B
55B	48B	43A
59B	2B	45B
23B	7B	50B
34B	16B	54A
10B	33B	57A
		61A
		65B
		62B
		5A
		3B
		9B
		6B
		15A
		9A
		14B
		20B
		22A
		29B
		27B

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4.5.3 Extreme temperature tests.- The following tests shall determine compliance with 3.5.

4.5.3.1 True state - (driver not loaded).

(a) Remove all previous external connections and apply power.

(b) Connect the following pins to power ground: 4B, 6A, 8B, 10A, 11B, 13A, 15B, 17A, 18B, 20A, 22B, 24A, 25B, 27A, 39B, 41A, 42B, 44A, 46B, 48A, 49B, 51A, 53B, 55A, 56B, 58B, 60B, 62A.

(c) Remove the connections to power ground from the pins in Column A of Table VIII, one at a time, and connect these pins to minus 5.58 volts plus or minus 2 percent each time removing the connection to minus 5.58 volts and replacing the connection to power ground before proceeding to the next row.

(d) While the pins in Column A of Table VIII are connected to minus 5.58 volts, read the voltage from the corresponding pin in Column B of Table V to power ground. This voltage shall be less than 50 volts.

4.5.3.2 True state - (driver loaded).

(a) Repeat 4.5.3.1 (a), (b), and (c).

(b) While the pins in Column A are connected to minus 5.58 volts, connect a 300K, 5%, 1/4w resistor from the corresponding pin in Column B to 139.5 volts plus or minus 2 percent. Read the voltage from the pin in Column B to power ground. This voltage shall be less than 80 volts.

TABLE VIII

A	B
B4, A6, B25, A27	A1
B4, A6, B11, A13	A2
B8, A10, B15, A17	A3
B8, A10, B25, A27	A7
B8, A10, B11, A13	A11
B18, A20, B15, A17	A14
B18, A20, B25, A27	A18
B18, A20, B11, A13	A21
B22, A24, B15, A17	A25
B22, A24, B25, A27	A28
B22, A24, B11, A13	A29
B46, A48, B42, A44	A38
B39, A41, B42, A44	A42
B53, A55, B42, A44	A45
B46, A48, B49, A51	A49
B53, A55, B49, A51	A52
B46, A48, B56, B58	A56
B39, A41, B56, B48	A59
B53, A55, B56, B58	A63
B46, A48, B60, A62	A64
B39, A41, B60, A62	A65

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4.5.3.3 False state.

(a) Repeat 4.5.3.1 (a) and (b).

(b) Remove the connections to power ground from the pins in Column A of table IX one row at a time, and connect these pins to minus 6.42 volts plus or minus 2% each time removing the connections to minus 5.58 volts and replacing the connections to power ground before proceeding to the next row.

(c) While the pins in Column A of Table IX are connected to minus 5.58 volts, read the voltage from the corresponding pin in Column B of Table IX to power ground. This voltage shall be greater than 77 volts.

TABLE IX

A	B
B4, A6, B25	A1
A6, B25, A27	A1
B25, A27, B4	A1
A27, B4, A6	A1
B4, A6, B11	A2
A13, B4, A6	A2
B11, A13, B4	A2
A6, B11, A13	A2
B8, A10, B15	A3
A17, B8, A10	A3
B15, A17, B8	A3
A10, B15, A17	A3
B8, A10, B25	A7
A27, B8, A10	A7
A25, A27, B8	A7
A10, B25, A27	A7
B8, A10, B11	A11
A13, B8, A10	A11
B11, A13, B8	A11
A10, B11, A13	A11
B18, A20, B15	A14
A17, B18, A20	A14
B15, A17, B18	A14
A20, B15, A17	A14
B18, A20, B25	A18
A27, B18, A20	A18
B25, A27, B18	A18
A20, B25, A27	A18
B18, A20, B11	A21
A13, B18, A20	A21
B11, A13, B18	A21
A20, B11, A13	A21

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TABLE IX (Cont'd)

A	B
B22, A24, B15	A25
A17, B22, A24	A25
B15, A17, B22	A25
A24, B15, A17	A25
B22, A24, B25	A28
A27, B22, A24	A28
B25, A27, B22	A28
A24, B25, A27	A28
B22, A24, B11	A29
A13, B27, A24	A29
B11, A13, B22	A29
A24, B11, A13	A29
B46, A48, B42	A38
A44, B46, A48	A38
B42, A44, B46	A38
A48, B42, A44	A38
B39, A41, B42	A42
A44, B39, A41	A42
B42, A44, B39	A42
A41, B42, A44	A42
B53, A55, B42	A45
A44, B53, A55	A45
B42, A44, B53	A45
A55, B42, A44	A45
B46, A48, B49	A49
A51, B46, A48	A49
B49, A51, B46	A49
A48, B49, A51	A49
B53, A55, B49	A52
A51, B53, A55	A52
B49, A51, B53	A52
A55, A49, A51	A52
B46, A48, B56	A56
B58, B46, A48	A56
B56, B58, A46	A56
A48, B56, B58	A56
B39, A41, B56	A59
B58, B39, A41	A59
B56, B58, B39	A59
A41, B56, B58	A59
B53, A55, B56	A63
B58, B53, A55	A63
B56, B58, B53	A63
A55, B56, B58	A63
B46, A48, B60	A64
A62, B46, A48	A64
B60, A62, B46	A64
A48, B60, A62	A64

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TABLE IX (Cont'd)

A	B
B39, A41, B60	A65
A62, B39, A41	A65
B60, A62, B39	A65
A41, B60, A62	A65

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing.- Preservation, packaging and packing shall be in accordance with Master Packing Data Sheet M14002.

6. NOTES

6.1 Intended use.- The readout driver is used to convert binary data to decimal form and driver numerical display tubes.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable level of preservation, packaging and packing.
(c) Applicable stock number.
(d) A block of serial numbers to identify each circuit board.

6.3 Design information.- Information regarding specific design not covered or referenced by this specification may be obtained from the procuring activity upon the specific request of the contractor.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1220-A193Y

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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